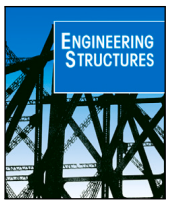
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Numerical analysis of bonded and unbonded prestressed RC beams using cohesive and non-compatible rod elements

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ABSTRACT

This research presents a numerical approach to simulate prestressed RC beams. It combines state-of-the-art finite element models to simulate the concrete (including fracture), tendons, passive reinforcements, and the rebar–concrete interaction. This provides more realistic simulations and rich details on deformability, crack patterns, and axial and contact stress distributions in passive reinforcements and tendons. Concrete fracture is simulated using cohesive elements with a constitutive model based on the plasticity theory and fracture mechanics, enabling crack development and propagation prediction. Reinforcements are simulated using rod elements with incompatible nodes. Special contact elements link rod to bulk elements and are placed along the reinforcements and at tips. These contact elements also predict relative displacements between rebars and concrete. Prestressing is simulated by setting initial stresses in tendons and applying forces to its nodes simultaneously. The approach also includes two numerical techniques to improve the convergence of the highly nonlinear problem. Five experimental beams are simulated. In all cases, the numerical load–deflection curves agree with the experiments. Fracture patterns also match rupture modes. Finally, contact stress discontinuities in tendons due to crack development are examined in more detail.

1. Introduction

Modeling RC (reinforced concrete) prestressed structures using finite elements requires carefully selecting elements and constitutive models to properly represent the mechanical behavior of the different materials and interactions. First, the concrete is a quasi-brittle material that, after cracking, presents an orthotropic nonlinear behavior due to the development of discontinuities. Additionally, studying the interaction between reinforcing bars (rebars) and concrete requires specific contact models and experimental parameters. Modeling rebars bearing tips also requires additional contact rules. As a bibliography review, Table 1 lists publications on the finite element simulation of prestressed RC beams. As can be seen, different approaches and simplifications were employed. For instance, some studies use isotropic models [1–6] to simulate the concrete and do not consider the development of cracks. Furthermore, as for the bond–slip behavior, some works do not simulate the bond conditions assuming perfect bonding implicitly. Similarly, some publications simplify the prestressing stage by applying temperature loads to represent tendon contraction [3,5,7] or replacing tendons with equivalent loads [8,9]. While these simplifications are reasonable given the progressive development of RC simulation techniques, many studies are limited to load–deflection curves, failing to capture important details such as crack patterns and distributions of axial and contact forces in rebars and tendons. This research combines different material models and finite elements to more accurately represent and simulate the mechanical behavior of prestressed RC beams.

Regarding concrete modeling, the literature includes several finite element models based on continuous and discontinuous approaches. On the continuous approach, the cracks are smeared and distributed over finite elements; see e.g. [10–17]. Isotropic and anisotropic damage models are also examples of smeared crack models [17–22]. Nevertheless, the continuous approach cannot model individual discontinuities, working better when cracks are distributed regularly throughout the material. On the other hand, in the discontinuous approach, cracks are modeled as displacement discontinuities between finite elements; see e.g. [23–47]. This feature is possible due to the addition of cohesive elements between neighbor bulk elements allowing for more realistic crack representations. The choice of the constitutive model is essential for an accurate representation of the cracks. For example, some plasticity-based models [28,36,40,46–48] can be employed along with cohesive elements. Within these models, the cohesive zone model (CZM) [49–51] is used to describe the relationship between stresses and crack openings to identify the development of new cracks.

Cohesive elements have been widely employed in the study of delamination and detachment of layered structures, with several studies reported in the literature [52–60]. Recently, cohesive elements

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Nomenclature

A_s	Rebar section area
$\tilde{\boldsymbol{B}}$	Strain–displacements matrix of a rod element
$\boldsymbol{D}_e$	Constitutive matrix for the elastic regime
$\boldsymbol{D}_{ep}$	Constitutive matrix for the elastoplastic regime
$\tilde{\boldsymbol{D}}$	Rebar–concrete element constitutive matrix
E	Young's modulus
E_s	Young's modulus of steel
f	Yield function
f_c	Concrete uniaxial compressive strength
f_t	Concrete uniaxial tensile strength
g	Plastic potential function
$\boldsymbol{f}$	Nodal forces vector
$\tilde{\boldsymbol{f}}$	Nodal forces of a rebar–concrete element
$\check{\boldsymbol{f}}$	Nodal forces at control points in a bulk elem.
$\tilde{\boldsymbol{f}}_{tip}$	Nodal forces of a tip-bearing element
G	Shear modulus
G_F	Fracture energy
h	Representative length of the two bulk elements that surround a cohesive element
$\boldsymbol{I}$	Identity matrix
J	Jacobian determinant
$\boldsymbol{J}$	Jacobian matrix
k_n	Normal stress *vs* normal relative displacement ratio
k_t	Tangential stress *vs* tangential relative displacement ratio
$\boldsymbol{K}$	Stiffness matrix
$\tilde{\boldsymbol{K}}$	Curved rod element stiffness matrix
$\tilde{\boldsymbol{K}}$	Rebar–concrete element stiffness matrix
$\tilde{\boldsymbol{K}}_{tip}$	Tip-bearing element stiffness matrix
ℓ	Longitudinal coordinate
$\boldsymbol{M}$	Interpolation matrix evaluated at control points of a bulk element
$\boldsymbol{N}$	Shape functions vector for a bulk element
$\tilde{\boldsymbol{N}}$	Shape functions vector for a rod element
$\boldsymbol{q}$	Vector of contact forces per unit length
$\boldsymbol{r}$	Vector of derivatives of the plastic potential function w.r.t. the stress components
$\boldsymbol{R}$	Rod element rotation matrix
s	Rebar slip
s_{peak}	Rebar slip corresponding to the peak contact stress
s_{res}	Rebar slip corresponding to the residual contact stress
T	Analysis pseudo-time
$\hat{\boldsymbol{t}}$	Unit vector computed at the rod tip
u_p	Effective plastic relative displacement
u_c	Critical crack opening
$\bar{u}$	Longitudinal displacement
$\boldsymbol{u}$	Nodal displacement vector of a bulk element
$\bar{\boldsymbol{u}}$	Nodal displacements of a rod element
$\tilde{\boldsymbol{u}}$	Nodal displacements of a rebar–concrete contact element
$\check{\boldsymbol{u}}$	Vector of displacements at control points in a bulk element
$\tilde{\boldsymbol{u}}_{tip}$	Nodal displacements of a tip-bearing element
$\boldsymbol{v}$	Vector of derivatives of the yield function w.r.t. the stress components
v	Tangential relative displacement
$\boldsymbol{w}$	Vector of relative displacements
$\boldsymbol{w}_e$	Vector of elastic relative displacements
$\boldsymbol{w}_p$	Vector of plastic relative displacements
$\boldsymbol{w}_{tip}$	Relative displacements vector of a tip-bearing element
w	Normal relative displacement
$\bar{w}_1$	Longitudinal relative displacement in a rebar–concrete contact element
$\bar{w}_2$	Relative displacement perpendicular to the longitudinal direction
α	Parameter that controls the yield function shape in the fracture model
β	Function to modify the size of the yield function in the fracture model
β_{ini}	Initial value of the β function
β_{res}	Residual value of the β function
$\bar{\varepsilon}$	Axial strain in a rod element
ζ	Factor that controls the relative elastic displacements for cohesive elements
θ	Parameter that defines the yield function reduction in the fracture model
$\dot{\lambda}$	Plastic multiplier
σ_n	Normal stress component
σ_{max}	Maximum value of the tensile stress recorded
$\boldsymbol{\sigma}$	Vector containing the normal and shear stress components
$\boldsymbol{\sigma}^{tr}$	Stress trial vector
$\boldsymbol{\sigma}^{n}$	Stress vector at step n
τ	Shear stress component
τ_{max}	Maximum contact stress
τ_{res}	Residual value of contact stress
τ^{tr}	Contact stress trial value

have gained interest for numerically simulating RC members, with applications including the modeling of fiber-reinforced concrete structures and rebar–concrete interactions [61,62]. Furthermore, cohesive elements have been coupled with reinforcement modeling to study degradation and crack properties in RC beams [63,64]. In particular, Maio et al. [63] developed a cohesive fracture model for predicting crack spacing and width in RC structures, demonstrating the potential of cohesive elements in capturing realistic fracture behavior in RC members.

The modeling of rebars is still problematic using conventional FE codes. The first problem is defining a finite element mesh compatible with the reinforcement layout. Another problem is defining a proper model for rebar–concrete interaction. When rod elements are used to simulate rebars, three methods are frequently described in the literature; herein, they are named *discrete*, *distributed*, and *embedded*. The discrete method represents the conventional approach of linking rod elements directly to bulk elements nodes or using contact elements as springs to represent bondings. In the distributed method, rebars are smeared over bulk elements considering rebar's inclination, and perfect bonding is implicitly assumed. The equivalent stiffness of the resulting composite is obtained by modifying the mechanical properties of the bulk element. This method is helpful for the analysis of reinforced structures with densely and uniformly distributed reinforcement. In the embedded method [65–68], rebars nodes are incompatible (Fig. 1), making the mesh generation process easier. However, the rebar stiffness is embedded within the trespassed bulk elements, thus limiting the application of boundary conditions such as force and displacements at the rebar nodes. One additional approach presented by Durand

Table 1
Finite element studies and modeling features on prestressing concrete beams.

Reference	Tendon bond	Concrete	Reinforcement/tendon	Bond–slip	Prestressing
Padmarajaiah et al. [1]	Bonded	Willam–Warnke plasticity model	Truss elements	Spring elements	By modifying the failure surface of concrete and the stress–strain response
Lou & Xiang [8]	Unbonded	1D beam element with nonlinear behavior	Equivalent nodal loads	None	Equivalent nodal loads
Mercan et al. [2]	Bonded	Concrete damage–plasticity	Truss elements	By modifying the post-peak behavior of concrete	By applying initial stresses
Ayoub et al. [71]	Bonded	Uniaxial law used based on Kent and Park model	Truss elements	Nonlinear bond elements	Force application at tendon ending nodes
Nie et al. [72]	Unbonded	Shell elements with uniaxial stress–strain curve and smeared crack modeling	Truss elements with uniaxial nonlinear models	Nonlinear spring elements	Force application at tendon ending nodes
You et al. [73]	Bonded	Smeared crack	Embedded bar elements	Perfect bond	Force application at ending nodes
Rabczuk et al. [7]	Bonded	Smeared crack in tension and non-local damage in compression	Beam elements with elastoplastic modeling	Interface elements	By applying temperature drop
Lou et al. [9]	Unbonded	1D nonlinear beam elements	Equivalent nodal loads	None	Equivalent nodal forces
Deng and Morcous [3]	Bonded	Willam–Warnke plasticity model	Truss elements with elastic-perfectly plastic modeling	Nonlinear spring elements	By applying temperature drop
Yapar et al. [4]	Bonded	Concrete damage-plasticity	Rod elements	Nonlinear spring elements	Force application at tendon ending nodes
Moreira et al. [74]	Unbonded	1D nonlinear frame element	Embedded polygonal element	None	Equivalent nodal forces
Thoma [75]	Bonded	Plate and shell elements with a cracked membrane model	Considered in the cracked membrane model	Shear-slip relation within the cracked membrane model	External loads
Brenkus et al [5]	Both	Concrete damage-plasticity modeling	Truss elements	Spring elements	Initial stresses or temperature drop
Pham and Hong [6]	Both	Concrete damage-plasticity modeling	Solid elements	By using frictional constraints	Prestressing forces at tendons
This research	Both	Cohesive elements with a plasticity-based fracture model	Rod elements with non-compatible nodes and nonlinear modeling	Rebar–concrete joint elements with nonlinear modeling	Initial stress at tendons, followed by the application of corresponding nodal forces

et al. [62] is the use of non-compatible rod elements linked to bulk elements by special contact elements (Fig. 1). Besides facilitating the mesh generation, the use of contact elements allows the simulation of different bond conditions, the application of boundary conditions at rod elements, and the use of tip-bearing elements. Fig. 2 shows a bulk and a rod element combined with contact elements. To model bond–slip behavior, there is a range of models available in the literature, see e.g. [62,69,70], and can be used along with contact elements.

This research combines state-of-the-art finite element models to simulate the concrete (including fracture), tendons, passive reinforcements,

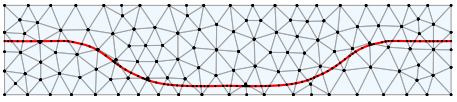


Fig. 1. Finite element mesh featuring rod elements with incompatible nodes.

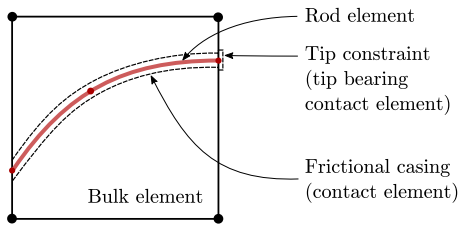


Fig. 2. An incompatible rod element linked to a bulk element through a contact element.

and the rebar–concrete contact to provide more realistic simulations and rich details about deformability, crack patterns, and axial and contact stress distributions in passive reinforcements and tendons. To simulate plain concrete, we propose a constitutive model for cohesive elements, which is formulated based on the plasticity theory and fracture mechanics. This model allows the simulation of the development and propagation of cracks, considering the concrete as an orthotropic material. The reinforcements are simulated using non-compatible rod elements. That is, rod elements are placed over the finite element mesh without the need for nodal compatibility with bulk elements. This flexibility is possible thanks to the addition of rebar–concrete contact elements capable of simulating the interaction between the two materials. Similarly, special contact elements are used to simulate the bearing tips of rebars. In turn, the prestressing stage simulation is performed by defining initial stresses at integration points of rod elements that represent the tendons and by simultaneously applying the resulting opposite forces to the corresponding nodes. Since the simulation of prestressed beams involves modeling concrete fracture and rebar–concrete bond failure, the resulting problem is highly nonlinear. Therefore, it cannot be easily solved by using a conventional solution procedure. For this reason, a second-order solution scheme is used to better approximate the global stiffness matrix for each loading increment. Also, an automatic loading scheme is used to increase the frequency of successful loading increments.

The paper is organized as follows. Section 2 presents details about the mesh generation process, including the generation of cohesive and rod elements and rebar–concrete contact elements. Section 3 introduces the constitutive modeling for plain concrete used in this work. Section 4 presents the formulation used to simulate the rebars, while Sections 5 and 6 present the formulation for the rebar–concrete contact and tip-bearing elements. Section 7 introduces the simulation procedure and techniques to improve the solution process. Finally, Section 8 presents the numerical simulation of prestressed RC, while Section 9 presents the conclusions.

2. Mesh generation

In the proposed approach to simulate prestressed concrete beams, the finite element mesh comprises five types of elements: bulk, cohesive joint, rod, rod–bulk contact, and tip-bearing elements. The concrete is simulated by bulk and zero-thickness cohesive elements. The bulk elements aim to simulate the concrete mass, while the cohesive elements simulate the development and propagation of cracks. Next, the reinforcement and tendons are simulated by non-compatible quadratic rod elements. In turn, the rebar–concrete contact is simulated by rod–bulk interface elements. Finally, the behavior of rebar endpoints as fixed or bearing ends is simulated by tip-bearing elements. The formulation and further details about these elements are described in the forthcoming sections.

The mesh generation steps are shown in Fig. 3. Initially, the domain is discretized as an unstructured mesh with a Delaunay mesh generator (see e.g. Ref. [76]). Next, cohesive elements are introduced between bulk elements using the algorithm presented by Durand and Silva [46].

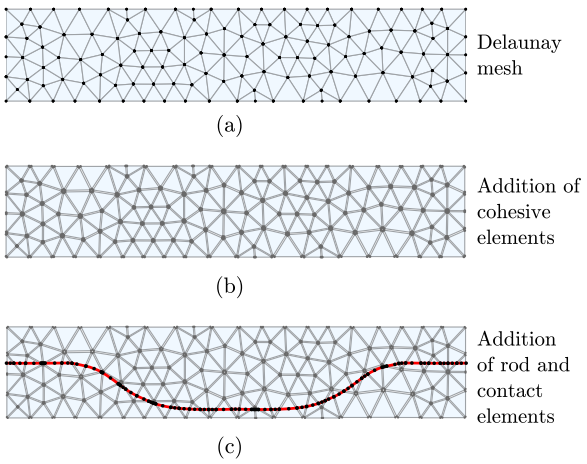


Fig. 3. Mesh generation process for the analysis of a RC beam.

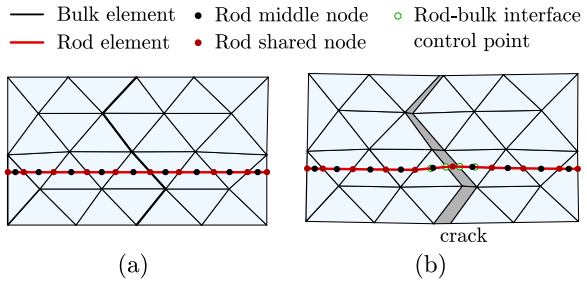


Fig. 4. Interaction between rod and cohesive elements: (a) undeformed mesh; (b) deformed mesh.

Finally, the rebars and tendons are discretized into rod elements using the algorithm presented by Durand et al. [77]. A reinforcement path may be defined by a polyline or any other mathematical curve; thus, curved reinforcements can be simulated. Individual rod elements simulate the rebar segments inside crossed bulk elements. The referenced algorithm includes the computation of intersection points between rebars and the boundaries of bulk elements. It is worth noting that all rod elements from a rebar are connected at end nodes. Three-node rod elements are used to obtain quadratic interpolation for geometry and displacements. Along with the generation of rod elements, special contact (rebar–concrete interface) elements [62] are also generated. Each contact element links a rod to the corresponding bulk element and can also be used to simulate different bond conditions. Furthermore, tip-bearing elements [62] are added at the ends of rebars to control the tip's relative displacement in the reinforcement direction. In a typical mesh of the proposed approach, reinforcement paths intersect with cohesive elements (Fig. 4a). Unlike bulk elements, rod elements are not detached when cracks develop; thus, no additional contact elements are required to link rod elements through cracks (Fig. 4b). The details about rebar–concrete contact elements will be presented in Section 5.

3. Concrete modeling

The numerical study of the mechanical behavior of quasi-brittle materials such as concrete requires an appropriate model to represent the deformability and the development of cracks. To this end, this research uses a continuous–discontinuous approach that combines linear bulk elements with nonlinear cohesive elements. The formulation of cohesive finite elements is well known (see e.g. Refs. [30,32,46,78–80]); for this reason, this section focuses on the formulation of the constitutive law.

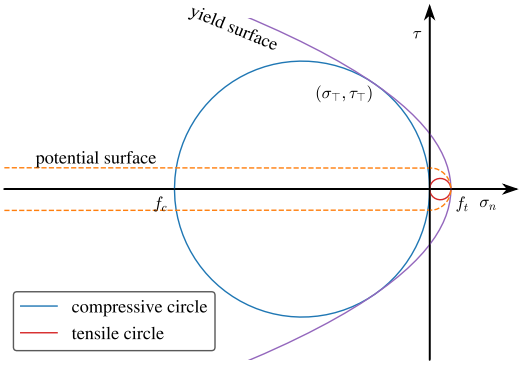


Fig. 5. Yield and potential plastic functions along with the compressive and tensile Mohr's circles.

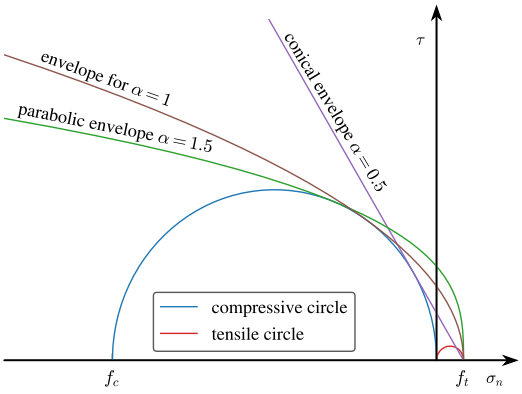


Fig. 6. Positive branch of yield curves for different values of α.

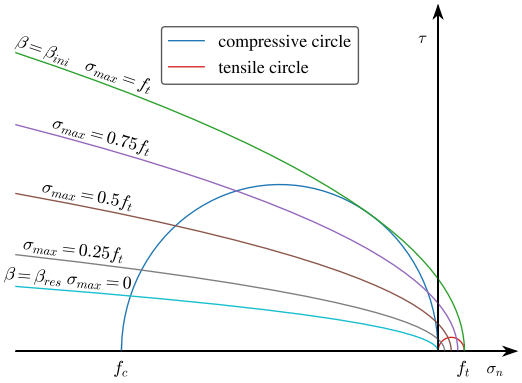


Fig. 7. Yield function evolution according to the maximum tensile strength σ_{max}.

3.1. Constitutive modeling

Cohesive elements require a reliable constitutive model that can accurately predict the formation and propagation of cracks, along with their associated openings. In this work, we propose a cohesive fracture model that builds upon a previous model developed by Durand et al. [46]. The key difference with the previous model is using a curved yield function that initially envelopes the tensile and compressive Mohr's circles, providing a more accurate representation of the material strength. Furthermore, the model accounts for strength reduction by moving the tip of the yield function and approximating the curve

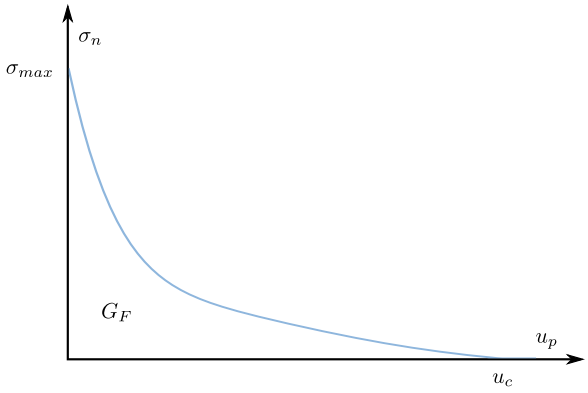


Fig. 8. Hordijk's traction–displacement curve σ_{max}.

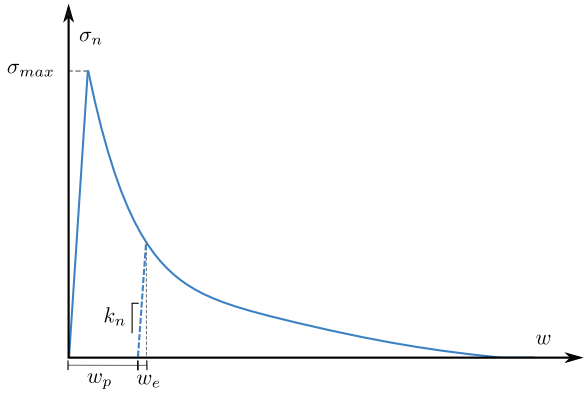


Fig. 9. Normal stress as a function of the crack opening.

branches. The proposed model is based on the plasticity theory and features a non-associated flow rule. The yield function (Eq. (1)) is defined as

$$f(\boldsymbol{\sigma}, \sigma_{max}) = \beta(\sigma_{max})(\sigma_n - \sigma_{max}) + \left(\frac{\tau^2}{f_t^2}\right)^{\alpha} = 0. \tag{1}$$

The maximum tensile strength σ_{max} recorded during the analysis is used as the internal softening variable. The exponent α is an input parameter that aims to provide a curved (exponential) yield function enveloping the tensile and compressive Morh circles characterized by the tensile f_t and compressive f_c strengths. In turn, β is used to modify the size of the yield function according to σ_{max} and is computed as

$$\beta(\sigma_{max}) = \beta_{res} + (\beta_{ini} - \beta_{res})\left(\frac{\sigma_{max}}{f_t}\right)^{\theta}, \tag{2}$$

where θ is a parameter that defines how fast the yield function is modified. The initial value for β is given by

$$\beta_{ini} = \frac{\left(\frac{\tau_\top^2}{f_t^2}\right)^{\alpha}}{f_t - \sigma_\top}, \tag{3}$$

where

$$\sigma_\top = \frac{2\alpha f_t + \alpha f_c - f_c - \sqrt{\alpha^2 f_c^2 - 4\alpha^2 f_c f_t + 4\alpha^2 f_t^2 - 2\alpha f_c^2 + f_c^2}}{4\alpha - 2}, \tag{4}$$

$$\tau_\top = \sqrt{\alpha(2\sigma_\top - f_c)(f_t - \sigma_\top)}. \tag{5}$$

The point $(\sigma_\top, \tau_\top)$ in Fig. 5 represents the intersection between the compressive Mohr's circle and the initial yield envelope. The constants $\sigma_\top$, $\tau_\top$ and β_{ini} are found after solving the following nonlinear system

$$\left(\sigma_\top - \frac{f_c}{2}\right)^2 + \tau_\top^2 = 0, \tag{6}$$

$$\beta_{ini}(\sigma_\tau - f_t) + \left(\frac{\tau_\tau^2}{f_t^2}\right)^\alpha = 0 \quad \text{and} \tag{7}$$

$$\left[\sigma_\tau - \frac{f_c}{2}, \tau_\tau\right]^T \left[-\frac{\partial f}{\partial \tau}, \frac{\partial f}{\partial \sigma_n}\right]\Big|_{\substack{\tau=\tau_\tau \\ \sigma_n=\sigma_\tau}} = 0. \tag{8}$$

The equations above represent, respectively, the compressive Mohr's circle evaluated at (σ_τ, τ_τ), the yield function evaluated at (σ_τ, τ_τ) and the normality condition between the compressive circle radius and the yield function slope at (σ_τ, τ_τ). Finally, the term β_{res} in Eq. (1) is the residual value for β which is assumed as a fraction of β_{ini}, e.g. $\beta_{res} = \gamma\,\beta_{ini}$. In the numerical simulations, γ was assumed as 0.01.

Fig. 6 displays different shapes of the positive branch according to the value of the α exponent. Finally, Fig. 7 shows the yield function evolution according to the reduction of the maximum tensile strength σ_{max}. On the other hand, the plastic potential function (see Fig. 5) is given by the composite function

$$g(\boldsymbol{\sigma}, \sigma_{max}) = \begin{cases} \tau^2 - \sigma_{max}^2 = 0 & \text{for} \quad \sigma_n \le 0 \\ \sigma_n^2 + \tau^2 - \sigma_{max}^2 = 0 & \text{for} \quad \sigma_n > 0 \end{cases} \tag{9}$$

The internal variable $\sigma_{max} = \sigma_{max}(u_p)$ defines the location of the yield surface vertex and aims to simulate the loss of strength due to cracking. This variable depends on the effective plastic relative displacement defined as

$$u_p = \int_0^t \|\dot{\boldsymbol{w}}_p\|\, dt = \int_0^t \sqrt{\dot{w}_p^2 + \dot{v}_p^2}\, dt, \tag{10}$$

where $\dot{w}_p$ and $\dot{v}_p$ are the rates of normal and shear plastic relative displacements, respectively, and $\dot{\boldsymbol{w}}_p = \langle \dot{w}_p, \dot{v}_p \rangle$.

Traction-displacement curves provide the relationship between σ_{max} and u_p. In this work, we use the curve proposed by Hordijk [81] (Fig. 8) which depends only on two parameters: $f_t = \sigma_{max}^0$ and the critical opening u_c. The Hordijk's softening curve, using the coefficients presented by Zivaljic et al. [82] is written as

$$\sigma_{max}(u_p) = \begin{cases} \left[\left(1 + 27\left(\frac{u_p}{u_c}\right)^3\right) e^{-6.93\frac{u_p}{u_c}} - 28\,\frac{u_p}{u_c}\, e^{-6.93}\right] f_t & \text{for} \quad u_p \le u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases}. \tag{11}$$

For this curve, the fracture energy can be expressed as a function of the tensile strength and the critical opening as $G_F = 0.1943 f_t u_c$. To illustrate how the normal stress behaves in relation to crack opening, we can refer to Fig. 9. This figure also features an unloading branch showcasing elastic and plastic displacement components.

3.2. Constitutive matrix

For the elastic regime, the constitutive matrix is given by

$$\boldsymbol{D}_e = \begin{bmatrix} k_n & 0 \\ 0 & k_t \end{bmatrix}, \tag{12}$$

where k_n and k_t relate, respectively, normal and tangential contact stresses to normal and tangential relative displacements. These coefficients are defined as

$$k_n = E\frac{\zeta}{h} \quad \text{and} \quad k_t = G\frac{\zeta}{h}, \tag{13}$$

where h is the representative length of bulk elements, E is the Young's modulus, G is the shear modulus and ζ is a factor that controls the elastic relative displacements. Since introducing cohesive elements into a finite element mesh increases the overall deformability [83,84], the calibration of ξ and h aims to minimize undesired elastic displacements as well as reduce mesh dependency. Most studies in this field typically take a single value of h as a representative value for the entire mesh. Furthermore, some studies have investigated the calibration of this value (see e.g. [84]), as excessively high values can lead to spurious traction oscillations due to the ill-conditioning of the resulting numerical problem. In contrast, this research employs a more localized approach. Each cohesive element is assigned a h value corresponding to the representative length of the two adjacent bulk elements, as described by Durand et al. [46]. This length is calculated for each cohesive element during a preprocessing step and allows us to set the elastic displacement factor ζ independently of the mesh density and local refinements. Nevertheless, this factor requires calibration to achieve minimal, negligible relative displacements within cohesive elements in the elastic regime.

For the elastoplastic regime, the constitutive matrix $\boldsymbol{D}_{ep}$ is formulated based on the conventional plasticity theory. According to the additive decomposition, the increment of relative displacements at a point is expressed as the sum of the elastic and plastic components

$$\dot{\boldsymbol{w}} = \dot{\boldsymbol{w}}_e + \dot{\boldsymbol{w}}_p. \tag{14}$$

The flow rule defines the increment of plastic relative displacements as

$$\dot{\boldsymbol{w}}_p = \dot{\lambda}\,\boldsymbol{r}, \tag{15}$$

where $\boldsymbol{r} = \frac{\partial g}{\partial \boldsymbol{\sigma}}$ is a vector normal to the potential surface g and $\dot{\lambda}$ is the plastic multiplier. Thus, the increment of the effective plastic relative displacement is given by

$$\dot{u}_p = \|\dot{\boldsymbol{w}}_p\| = \dot{\lambda}\,\|\boldsymbol{r}\|. \tag{16}$$

Following the consistency condition, the plastic multiplier is given by

$$\dot{\lambda} = \frac{\boldsymbol{v} \cdot \boldsymbol{D}_e\, \dot{\boldsymbol{w}}}{\boldsymbol{v} \cdot \boldsymbol{D}_e\, \boldsymbol{r} - \frac{\partial f}{\partial \sigma_{max}} \frac{\partial \sigma_{max}}{\partial u_p} \|\boldsymbol{r}\|}, \tag{17}$$

where $\boldsymbol{v} = \frac{\partial f}{\partial \boldsymbol{\sigma}}$. Finally, the elastoplastic constitutive matrix $\boldsymbol{D}_{ep}$ is computed as

$$\boldsymbol{D}_{ep} = \boldsymbol{D}_e - \frac{\boldsymbol{D}_e\, \boldsymbol{r}\, \boldsymbol{v}^T\, \boldsymbol{D}_e}{\boldsymbol{v}^T\, \boldsymbol{D}_e\, \boldsymbol{r} - \frac{\partial f}{\partial \sigma_{max}} \frac{\partial \sigma_{max}}{\partial u_p} \|\boldsymbol{r}\|}. \tag{18}$$

3.3. Stress update

Given the increment of relative displacements at an integration point, the stress update process begins by computing the trial stress vector:

$$\boldsymbol{\sigma}^{tr} = \boldsymbol{\sigma}^n + \boldsymbol{D}_e\, \Delta \boldsymbol{w}, \tag{19}$$

where the superscript n represents the current step. Then, the yield function is evaluated for $\boldsymbol{\sigma}^{tr}$ to find whether the current increment is elastic or elastoplastic. For elastic increments including unloading/reloading conditions, $\boldsymbol{\sigma}^{n+1}$ is considered equal to $\boldsymbol{\sigma}^{tr}$. For elastoplastic loadings, the internal variable σ_{max} decreases and the yield function shape is modified. Then, the stress state, initially approximated by $\boldsymbol{\sigma}^{tr}$, returns to the yield surface following the flow rule, as shown in Fig. 10. The values of u_p, σ_{max} and $\boldsymbol{\sigma}$ at step $n+1$, as functions of the incremental value of the plastic multiplier $\Delta\lambda$, are given by

$$u_p^{n+1} = u_p^n + \Delta\lambda \|\boldsymbol{r}^{n+1}\|, \tag{20}$$

$$\sigma_{max}^{n+1} = \sigma_{max}^n + \frac{\partial \sigma_{max}}{\partial u_p}\, \Delta\lambda \|\boldsymbol{r}^{n+1}\|, \tag{21}$$

$$\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{tr} - \Delta\lambda\, \boldsymbol{D}_e\, \boldsymbol{r}^{n+1}. \tag{22}$$

After substituting the above equations into the consistency condition $f(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1}) = 0$, $\Delta\lambda$ can be found numerically, e.g. using the Newton–Raphson method. Hence, the stress state and the values of the internal variables can be found at step $n+1$.

4. Reinforcement simulation

Reinforcements such as tendons and passive steel are simulated using quadratic rod elements and the well-known one-dimensional elastic perfectly plastic constitutive model. This section briefly describes the formulation for curved rod elements used in this research.

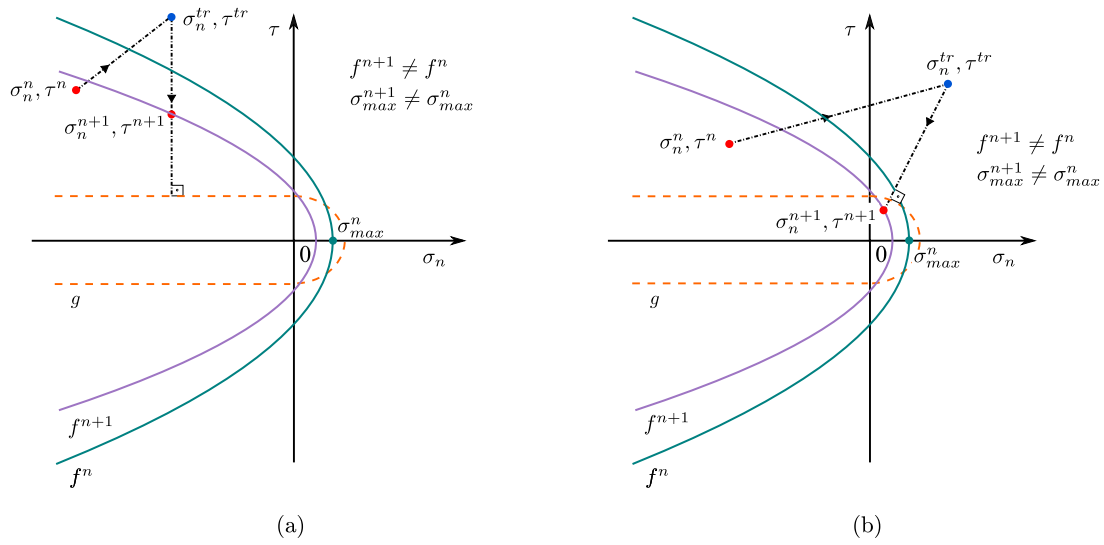


Fig. 10. Stress update in the elastoplastic regime: (a) stress return to the yield surface when $\sigma_n^{tr} < 0$; (b) stress return to the yield surface when $\sigma_n^{tr} > 0$.

4.1. Curved rod element

The strain at a point along a rod element is defined as $\bar{\varepsilon} = \frac{d\bar{u}}{d\ell}$, where $\bar{u}$ is the longitudinal displacement, and ℓ is the longitudinal coordinate. In a curved rod, the displacement $\bar{u}$ can be computed by interpolation of nodal displacements as

$$\bar{u} = \hat{\boldsymbol{J}} \cdot \left(\bar{\boldsymbol{N}}\bar{\boldsymbol{u}}\right), \tag{23}$$

where $\hat{\boldsymbol{J}}$ is the Jacobian unit vector, and $\bar{\boldsymbol{N}}$ is given by the matrix

$$\bar{\boldsymbol{N}} = \begin{bmatrix} \bar{N}_1\boldsymbol{I} & \bar{N}_2\boldsymbol{I} & \bar{N}_3\boldsymbol{I} & \cdots & \bar{N}_{\bar{n}}\boldsymbol{I} \end{bmatrix}, \tag{24}$$

where $\bar{n}$ is the rod's number of nodes and $\boldsymbol{I}$ is a 2×2 identity matrix. By computing $\frac{\partial \bar{u}}{\partial \ell}$, the axial strain is obtained as

$$\varepsilon = \frac{\boldsymbol{J}^T}{\boldsymbol{J}^2}\frac{d\bar{\boldsymbol{N}}}{d\bar{\xi}}\bar{\boldsymbol{u}}. \tag{25}$$

where $\bar{\boldsymbol{B}} = \frac{\boldsymbol{J}^T}{\boldsymbol{J}^2}\frac{d\bar{\boldsymbol{N}}}{d\bar{\xi}}$ is a matrix that relates the axial strain to the rod nodal displacements. Afterward, the internal forces can be computed as

$$\bar{\boldsymbol{f}} = \int_{\ell} \bar{\boldsymbol{B}}^T \bar{\sigma} A_s \, \mathrm{d}\ell, \tag{26}$$

where $\bar{\sigma}$ is the axial stress and A_s is the rod section area.

4.2. Stiffness matrix

The stiffness matrix of a rod element is found by computing the derivative $\frac{d\bar{f}}{d\bar{u}}$ (see Eq. (26)). To this end, the derivative $\frac{d\bar{\sigma}}{d\bar{u}}$ is obtained as

$$\frac{d\bar{\sigma}}{d\bar{\boldsymbol{u}}} = \frac{d\bar{\sigma}}{d\bar{\varepsilon}}\frac{d\bar{\varepsilon}}{d\bar{\boldsymbol{u}}} = E_s\bar{\boldsymbol{B}}, \tag{27}$$

where E_s is the rebar tangent stiffness and relates the axial stress to the axial strain. Then, substituting Eq. (27) into $\frac{d\bar{f}}{d\bar{u}}$, the stiffness matrix is found as

$$\bar{\boldsymbol{K}} = \int_{\ell} \bar{\boldsymbol{B}}^T \bar{\boldsymbol{B}} \, E_s A_s \, \mathrm{d}\ell. \tag{28}$$

5. Rebar–concrete contact element

A rebar–concrete (or rod–bulk) contact element is used to link a reinforcing rod element to the surrounding bulk element (Fig. 2). Fig. 11 presents the nomenclature for nodal displacements of the bulk and rod element. The figure also shows the displacements of control points located at the bulk element side. These control points are used to compute the relative displacement between the bulk and rod elements. The connectivity of a rod–bulk contact element is given by the collection of bulk and rod elements nodes; thus, the vector of nodal displacements is given by

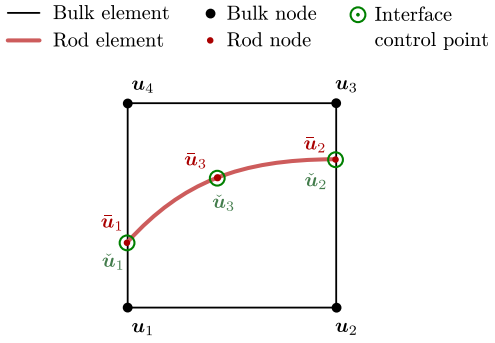


Fig. 11. Finite element representation of the bulk and rod elements. The nodes of a rod–bulk contact element are given by the collection of bulk and rod nodes.

$$\begin{aligned} \tilde{\boldsymbol{u}} &= \langle \boldsymbol{u}, \bar{\boldsymbol{u}} \rangle \\ &= \Big\langle \underbrace{u_{1x}, u_{1y}, u_{2x}, u_{2y}, \ldots, u_{nx}, u_{ny}}_{\boldsymbol{u}}, \\ &\quad \underbrace{\bar{u}_{1x}, \bar{u}_{1y}, \bar{u}_{2x}, \bar{u}_{2y}, \cdots \bar{u}_{\bar{n}x}, \bar{u}_{\bar{n}y}}_{\bar{\boldsymbol{u}}}, \Big\rangle. \end{aligned} \tag{29}$$

5.1. Relative displacements

Relative displacements between a bulk and rod element are computed using the rod nodal displacements $\bar{\boldsymbol{u}}$ and the displacements $\check{\boldsymbol{u}}$ at control points located at bulk elements; see Fig. 11. Thus, the relative displacements vector at rode node i is defined as

$$\boldsymbol{w}_i = \bar{\boldsymbol{u}}_i - \check{\boldsymbol{u}}_i. \tag{30}$$

The displacements $\check{\boldsymbol{u}}_i$ are computed by interpolation of the bulk element nodal displacements as

$$\check{\boldsymbol{u}}_i = \sum_m N_{m,i}\,\boldsymbol{u}_m, \tag{31}$$

where $N_{m,i}$ represents the mth shape function evaluated at the bulk local coordinate of the control point i. The displacement vector for all control points can be expressed as $\check{\boldsymbol{u}} = \boldsymbol{M}\boldsymbol{u}$, where

$$\boldsymbol{M} = \begin{bmatrix} N_{1,1}\boldsymbol{I} & N_{2,1}\boldsymbol{I} & N_{3,1}\boldsymbol{I} & \cdots & N_{n,1}\boldsymbol{I} \\ N_{1,2}\boldsymbol{I} & N_{2,2}\boldsymbol{I} & N_{3,2}\boldsymbol{I} & \cdots & N_{n,2}\boldsymbol{I} \\ N_{1,3}\boldsymbol{I} & N_{2,3}\boldsymbol{I} & N_{3,3}\boldsymbol{I} & \cdots & N_{n,3}\boldsymbol{I} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ N_{1,\bar{n}}\boldsymbol{I} & N_{2,\bar{n}}\boldsymbol{I} & N_{3,\bar{n}}\boldsymbol{I} & \cdots & N_{n,\bar{n}}\boldsymbol{I} \end{bmatrix}, \tag{32}$$

After computing the nodal relative displacements, the value at any point along the rod can be interpolated as

$$\boldsymbol{w}(s) \approx \sum_i \bar{N}_i(\bar{\xi})\,\boldsymbol{w}_i\,, \tag{33}$$

where $\bar{\xi}$ is the natural coordinate in the rod system. Next, the vector of relative displacements can be expressed as a function of the contact element displacement vector as

$$\bar{\boldsymbol{w}} = \boldsymbol{R}\begin{bmatrix} -\bar{\boldsymbol{N}}\boldsymbol{M} & \bar{\boldsymbol{N}} \end{bmatrix}\begin{Bmatrix} \boldsymbol{u} \\ \bar{\boldsymbol{u}} \end{Bmatrix}. \tag{34}$$

where $\boldsymbol{R} = \boldsymbol{R}(\bar{\xi})$ is the rebar rotation matrix (see Ref. [62]), $\bar{\boldsymbol{N}} = \bar{\boldsymbol{N}}(\bar{\xi})$ is given in Eq. (24) and matrix

$$\tilde{\boldsymbol{B}} = \boldsymbol{R}\begin{bmatrix} -\bar{\boldsymbol{N}}\boldsymbol{M} & \bar{\boldsymbol{N}} \end{bmatrix} \tag{35}$$

can be used to relate the bulk and rod displacements to relative displacements at a point $\bar{\xi}$ in the rod's local system.

5.2. Contact force

The contact forces are considered to be continuously distributed along the rod element. The rate of change in the contact force vector per unit length is composed of two orthonormal components

$$\dot{\boldsymbol{q}} = \dot{\tau}\,p\,\boldsymbol{e}_1 + \dot{q}_{n2}\,\boldsymbol{e}_2, \tag{36}$$

where $\tau(\bar{w}_1)$ is the contact shear stress, p is the perimeter of the rod's cross-section, $q_n(\bar{w}_2)$ is the distributed contact force orthogonal to the rod and $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ are the longitudinal and orthogonal directions, respectively. Assuming that the contact forces are functions of the relative displacements, we have

$$\dot{\boldsymbol{q}} = \dot{\bar{w}}_1 \frac{d\tau}{d\bar{w}_1}\,p\,\boldsymbol{e}_1 + \dot{\bar{w}}_2\,k_n\,p\,\boldsymbol{e}_2, \tag{37}$$

where $\frac{d\tau}{d\bar{w}_1}$ is the longitudinal tangent stiffness and k_n a penalty parameter chosen to provide negligible displacements in the orthogonal direction. Integrating $\boldsymbol{q}$ along the rod length, the forces at control points are found as

$$\check{\boldsymbol{f}} = \int_\ell \bar{\boldsymbol{N}}^T \boldsymbol{q}\,\mathrm{d}\ell. \tag{38}$$

Due to the equilibrium condition, the corresponding forces vector at rod nodes is given by $\bar{\boldsymbol{f}} = -\check{\boldsymbol{f}}$. In turn, the vector of corresponding nodal forces acting on the bulk element is obtained as

$$\boldsymbol{f} = \boldsymbol{N}^T \check{\boldsymbol{f}}. \tag{39}$$

5.3. Stiffness matrix

The stiffness matrix of the contact element is computed as $\tilde{\boldsymbol{K}} = \frac{d\tilde{\boldsymbol{f}}}{d\tilde{\boldsymbol{u}}}$, where $\tilde{\boldsymbol{f}} = \langle \boldsymbol{f}, \bar{\boldsymbol{f}} \rangle$; thus

$$\tilde{\boldsymbol{K}} = \begin{bmatrix} \frac{d\boldsymbol{f}}{d\boldsymbol{u}} & \frac{d\boldsymbol{f}}{d\bar{\boldsymbol{u}}} \\ \frac{d\bar{\boldsymbol{f}}}{d\boldsymbol{u}} & \frac{d\bar{\boldsymbol{f}}}{d\bar{\boldsymbol{u}}} \end{bmatrix}. \tag{40}$$

After computing the derivatives, the stiffness matrix can finally be expressed as

$$\tilde{\boldsymbol{K}} = \int_\ell \tilde{\boldsymbol{B}}^T \tilde{\boldsymbol{D}}\,\tilde{\boldsymbol{B}}\,p\,\mathrm{d}\ell. \tag{41}$$

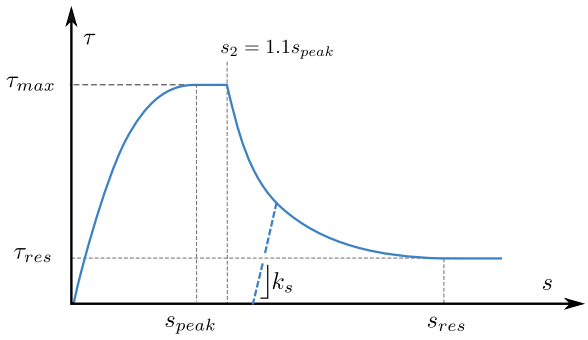


Fig. 12. Bond–slip model envelope.

where $\tilde{\boldsymbol{D}}$ is the contact constitutive matrix

$$\tilde{\boldsymbol{D}} = \begin{bmatrix} \frac{d\tau}{d\bar{w}_1} & 0 \\ 0 & k_n \end{bmatrix}. \tag{42}$$

The stiffness matrix in Eq. (41) can be approximated using numerical integration. At least two integration points are required for contact elements with linear interpolation.

5.4. Bond–slip constitutive model

The bond–slip model aims to provide the rate of change of the longitudinal contact stress with respect to the rebar slip, e.g. $\frac{d\tau}{ds}$, where $s = \bar{w}_1$. In this work, we use the model presented by Durand et al. [62], which in turn is based on the cyclic model presented by Murcia-Delso and Shing [70]. Here, a simplified version for monotonic loadings is presented (Fig. 12). In this model, the shear strength is given by the following piecewise function

$$\tau(s) = \begin{cases} \tau_{max}\left(\dfrac{s}{s_{peak}}\right)^{\alpha} & \text{for} \quad s \le s_{peak}, \\ \tau_{max} & \text{for} \quad s_{peak} < s \le 1.1 s_{peak}, \\ \tau_{max} - \left(\tau_{max} - \tau_{res}\right)\left(\dfrac{s - 1.1 s_{peak}}{s_{res} - 1.1 s_{peak}}\right)^{\beta} & \text{for} \quad 1.1 s_{peak} < s \le s_{res} \\ \tau_{res} & \text{for} \quad s_{res} < s \end{cases} \tag{43}$$

In the equation above, τ_{max} and τ_{res} are the maximum and residual contact strength, respectively, α and β are parameters that define the curvature of the ascending and descending branches, respectively, and s_{peak} and s_{res} are characteristic slip values according to Fig. 12.

The stress integration step in a finite element analysis proceeds as follows, First, for a given increment Δs at step n, the elastic trial stress is given by

$$\tau^{tr} = \tau^n + k_s\,\Delta s. \tag{44}$$

Considering $s^{n+1} = s^n + \Delta s$, a prediction for the stress envelope at s^{n+1} is computed according to the sign of $s^{n+1}\,\tau^{tr}$ as

$$\tau_{env}^{n+1} = \begin{cases} \tau_{res} & \text{for} \quad s^{n+1}\,\tau^{tr} < 0 \\ \max(\tau_{res}, \tau(|s^{n+1}|)) & \text{for} \quad s^{n+1}\,\tau^{tr} \ge 0, \end{cases} \tag{45}$$

where $\tau(|s^{n+1}|)$ is evaluated according to Eq. (43). Finally, the stress at step $n+1$ is computed as

$$\tau^{n+1} = \begin{cases} \tau^{tr} & \text{for} \quad |\tau^{tr}| < \tau_{env}^{n+1} \\ \tau_{env}^{n+1}\,\mathrm{sgn}\,(\tau^{tr}) & \text{for} \quad |\tau^{tr}| \ge \tau_{env}^{n+1} \end{cases} \tag{46}$$

It is worth mentioning that the bond–slip model does not explicitly consider the effects of the steel deformation; however, the resulting behavior presented in Fig. 12 is compatible with experimental pull-out tests (see e.g. Ref. [85]).

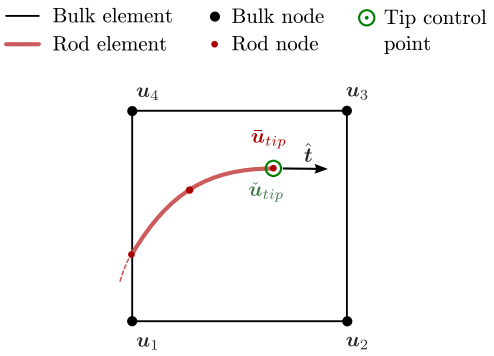


Fig. 13. Finite element representation showing the tip node and corresponding control point in the bulk element.

6. Tip-bearing element

Tip-bearing elements are placed at rebar ends and are used to restrain relative displacements in the longitudinal direction. The formulation presented here is a two-dimensional version of the element presented by Durand et al. [62]. Tip-bearing elements can be used to simulate two conditions: (1) fully restrained tips (anchored tips) and (2) tips restrained to avoid penetration into the reinforced medium (non-anchored tips). These features help simulate RC prestressed beams where the tendon ends are fixed and also avoid the penetration of passive reinforcement into concrete. The relative displacement at a tip is computed from the displacements of the tip node and the corresponding control point located inside the bulk element. Fig. 13 shows a finite element representation of the tip node and the corresponding control point. The nodes of a tip-bearing element are given by the collection of all nodes from the bulk element plus the rebar tip node. Thus, the displacement vector is given by

$$\tilde{\boldsymbol{u}}_{tip} = \langle \boldsymbol{u}, \bar{\boldsymbol{u}}_{tip} \rangle = \langle \underbrace{u_{1x}, u_{1y}, u_{2x}, u_{2y}, \dots, u_{nx}, u_{ny}}_{\boldsymbol{u}}, \underbrace{\bar{u}_{tip\,x}, \bar{u}_{tip\,y}}_{\bar{\boldsymbol{u}}_{tip}} \rangle \tag{47}$$

where $\bar{\boldsymbol{u}}_{tip}$ is a vector with the tip node displacements.

6.1. Stiffness matrix

The relative displacement between a tip node and the corresponding control point is defined as

$$\boldsymbol{w}_{tip} = \bar{\boldsymbol{u}}_{tip} - \check{\boldsymbol{u}}_{tip} \tag{48}$$

where $\bar{\boldsymbol{u}}_{tip}$ is a vector with the tip displacements and $\check{\boldsymbol{u}}_{tip}$ a vector with the displacements at the control point. The latter is computed by interpolation of the bulk element displacements as

$$\check{\boldsymbol{u}}_{tip} = \boldsymbol{M}_{tip}\boldsymbol{u}. \tag{49}$$

In the equation above, matrix $\boldsymbol{M}_{tip}$ is given by

$$\boldsymbol{M}_{tip} = \begin{bmatrix} N_{1,tip}\boldsymbol{I} & N_{2,tip}\boldsymbol{I} & N_{3,tip}\boldsymbol{I} & \cdots & N_{n,tip}\boldsymbol{I} \end{bmatrix}, \tag{50}$$

where $N_{m,tip}$ represents the mth shape function from the bulk element evaluated at the natural coordinates corresponding to the rod tip. Then, using Eqs. (48) and (49), the relative displacement at the tip in the rod longitudinal direction is computed as

$$\iota\bar{w}_{tip} = \hat{\boldsymbol{t}}^T \begin{bmatrix} -\boldsymbol{M}_{tip} & \boldsymbol{I} \end{bmatrix} \begin{Bmatrix} \boldsymbol{u} \\ \bar{\boldsymbol{u}}_{tip} \end{Bmatrix}, \tag{51}$$

where $\hat{\boldsymbol{t}}$ is a unit vector parallel to the rod computed at the tip location (see Fig. 13) and $\boldsymbol{B}_{tip} = \hat{\boldsymbol{t}}^T \begin{bmatrix} -\boldsymbol{M}_{tip} & \boldsymbol{I} \end{bmatrix}$ is a matrix that relates the nodal displacements $\tilde{\boldsymbol{u}}_{tip}$ from the tip-bearing element to the relative displacement $\iota\bar{w}_{tip}$ at the tip.

The internal forces vector of the tip-bearing element is computed as $\tilde{\boldsymbol{f}}_{tip} = \bar{f}_{tip}\boldsymbol{B}_{tip}^T$, where $\bar{f}_{tip}$ is the reaction force at the tip. Then, the stiffness matrix is computed as $\tilde{\boldsymbol{K}}_{tip} = \frac{d\tilde{\boldsymbol{f}}_{tip}}{d\tilde{\boldsymbol{u}}_{tip}}$. After computing the derivatives, the stiffness matrix is given by

$$\tilde{\boldsymbol{K}}_{tip} = \frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}\boldsymbol{B}_{tip}^T\boldsymbol{B}_{tip}, \tag{52}$$

where $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is the tangent stiffness that relates increments of force and relative displacement at the bearing tip. For anchored tips, $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is given by a penalty parameter k_{tip} in order to provide a negligible tip displacement. For non-anchored tips, $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ should be computed from a nonlinear force–displacement model.

6.2. Force–displacement model

The force–displacement model for a non-anchored tip-bearing element aims to provide a negligible tip displacement when simulating a rebar being pushed against the concrete. The auxiliary vector $\hat{\boldsymbol{t}}$ pointing outward at the rebar tip is used to identify if the rebar is being pushed or pulled. In an incremental analysis, the stiffness $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is computed as follows. If $\bar{w}_{tip} > 0$, $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}}$ is assumed equal to a penalty parameter k_{tip}; otherwise, $\frac{d\bar{f}_{tip}}{d\bar{w}_{tip}} = 0$. Then, the stiffness matrix is computed according to Eq. (52).

At the stress integration stage, given the increment $\Delta\bar{w}_{tip}$ at step n, the force increment at the tip is updated as follows. First, a trial value is computed as $f_{tip}^{tr} = f_{tip}^{n} + k_{tip}\Delta_w$, where f_{tip}^{n} is the tip force at step n. Next, if $f_{tip}^{tr} > 0$, the force is updated as $f_{tip}^{n+1} = f_{tip}^{tr}$. Otherwise, it is updated as $f_{tip}^{n+1} = 0$.

7. Simulation aspects

This section details the simulation aspects used to model RC prestressed beams. First, the procedure used to simulate the prestressing step is presented. Afterward, we present two techniques to improve the numerical convergence during the finite element analysis: (1) a Runge–Kutta-based method to improve the approximation of the global stiffness matrix; and (2) an automatic loading scheme. All implementations and numerical simulations were performed using the Amaru library (see Refs. [47,62] and https://github.com/NumSoftware/Amaru).

7.1. Prestressing forces

The numerical simulation of prestressed RC structures is usually performed in two stages (Fig. 14). The first one is dedicated to applying prestressing forces; thus, after this stage, the finite element model is mainly under compression. In the second stage, the loads are applied using an incremental scheme until failure.

In the proposed simulation approach, the finite element models comprise bulk, cohesive, rod, and contact elements. Initially, the tip-bearing elements at the tendon ending points are set as anchored. In unbonded tendons, the longitudinal contact parameter k_s is properly set to simulate low friction or sliding. Then, the prestressing forces are set through a combination of initial stresses and the application of load forces as follows. First, the tensile stresses corresponding to a tendon are set at the integration points of all rod elements representing that tendon (Fig. 14a). Then, the solver computes the internal forces for each rod and applies the opposite forces to the corresponding nodes (Fig. 14b). Since the rod is connected to the remaining RC structure, these forces will be redistributed in the beam. This effect represents the tendon's release as in an actual prestressing procedure. Finally, Fig. 14c shows an idealization of resulting forces applied to the prestressed beam.

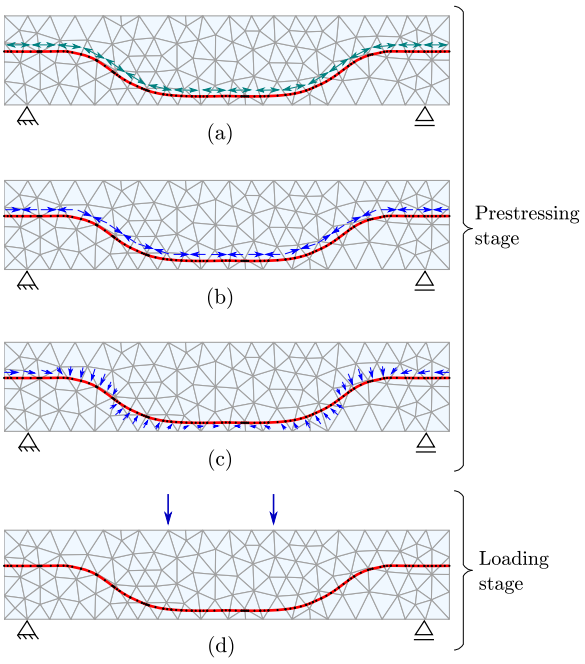


Fig. 14. Prestressing and loading stages: (a) setting initial stress at integration points of rod elements; (b) application of forces at rod nodes; (c) idealization of resulting forces acting on the beam; (d) loading.

7.2. Runge–Kutta estimative of the stiffness matrix

Because of the high nonlinearity of the problem (i.e., due to concrete cracking and rebar bond failure), conventional solution procedures tend to miss convergence in incremental analyses due to large residual forces. A conventional procedure to solve a nonlinear finite element problem frequently uses an incremental loading scheme combined with the Newton–Raphson method, approximating the global stiffness matrix at the beginning of each loading step. In order to improve this procedure, a second-order Runge–Kutta scheme called Ralston's method [86] is used to estimate a more representative stiffness matrix for each increment. To this end, initially, two stiffness matrices are computed as $\boldsymbol{K}_1 = \boldsymbol{K}_1(\boldsymbol{U}^n)$ and $\boldsymbol{K}_2 = \boldsymbol{K}_1(\boldsymbol{U}^n + \frac{2}{3}\Delta\boldsymbol{U})$, where $\boldsymbol{U}^n$ is the global displacements vector at the beginning of the step, and $\Delta\boldsymbol{U}$ is the displacements increment vector. Then, the final stiffness matrix is computed as

$$\bar{\boldsymbol{K}} = \frac{1}{4}\boldsymbol{K}_1 + \frac{3}{4}\boldsymbol{K}_2. \tag{53}$$

This procedure improves the success frequency of load increments by enhancing the results of displacements and internal forces and, consequently, reducing the residual forces.

7.3. Automatic loading stepping

In a nonlinear incremental analysis, the load and displacement increments at a given step are usually given by

$$\Delta\boldsymbol{F} = \Delta T\, \boldsymbol{F}_{ext} \quad \text{and} \tag{54}$$

$$\Delta\boldsymbol{U} = \Delta T\, \boldsymbol{U}, \tag{55}$$

where $T \in [0, 1]$ is the pseudo-time for the current analysis stage and ΔT is the step size used to scale the external forces $\boldsymbol{F}_{ext}$ and displacements $\boldsymbol{U}$ vectors. An automatic stepping algorithm modifies ΔT to increase after a successful increment ($err \leq tol$) and decrease when an increment trial fails. Thus, the increment size at step $n+1$ can be computed based on the previous increment as

$$\Delta T^{n+1} = q\, \Delta T^n, \tag{56}$$

Table 2
Author and mean features of the experimental beams.

Beam	Author	Prestressing	Tendon bond	Failure mode
B4	Hussien et al. [87]	–	–	A
B5	Hussien et al. [87]	Post-tensioned	Bonded	B
B9	Hussien et al. [87]	Post-tensioned	Unbonded	B
I	Eibl et al. [88]	Pre-tensioned	Bonded	B'
A-1	Tao and Du [89]	Post-tensioned	Unbonded	B'

(A) Yielding of steel reinforcement followed by crushing of concrete. (B) Yielding of non-prestressed steel followed by crushing of concrete then rupture of the prestressing tendon. (B') Yielding of non-prestressed steel followed by yielding of the prestressing tendon.

where q is a factor that aims to limit the truncation error defined here as $err = \|\Delta\boldsymbol{F}_{ext} - \Delta\boldsymbol{F}_{int}\|$. Following the expression proposed by Durand and Silva [47], in this research q is given by

$$q = 1 + \tanh\left(\log_{10}\frac{tol}{err}\right). \tag{57}$$

If the last increment trial converged successfully ($err \leq tol$), the increment size grows according to Eq. (56), otherwise the increment size decreases according to

$$\Delta T_{n+1} = \max(0.2,\ \min(0.9,\ q))\ \Delta T_n. \tag{58}$$

It is worth mentioning that in all stages, prestressing and loading, the loads are applied incrementally using this automatic stepping scheme. This scheme reduces simulation times by scaling the load increments as needed.

8. Numerical simulations

This section presents the numerical simulation of five concrete beams studied experimentally by Hussien et al. [87], Eibl et al. [88], and Tao and Du [89]. The main features of these beams are listed in Table 2 where the same nomenclature as in the original papers is used. All beams are prestressed except beam B4, studied as a reference for beams B5 and B9. Fig. 15 shows all the beams, including the passive reinforcements, tendons, loads, and supports.

In most cases, the material parameters for the constitutive models were obtained from the original papers. Missing parameters were estimated according to existing approximations. Hence, the tensile strength of concrete is estimated based on the recommendations of the CEB-FIP [90]. In this case, the mean tensile strength is computed first as

$$f_{tm} = \begin{cases} 0.3|f_c|^{2/3} & \text{for concrete grades } \leq \text{C50} \\ 2.12\ln\left(1 + 0.1 f_c\right) & \text{for concrete grades} > \text{C50}. \end{cases} \tag{59}$$

Then, the actual value for f_t is taken within the interval $0.7 f_{cm} \leq f_t \leq 1.3 f_{cm}$. Similarly, following the CEB-FIP [90] recommendations, the concrete fracture energy is estimated as

$$G_F = 73|f_c|^{0.18} \tag{60}$$

and the Young modulus as

$$E_c = 21500\,\alpha_E \left(\frac{f_c}{10}\right)^{1/3}, \tag{61}$$

where α_E depends on the type of aggregate and is adopted as 1.0 in this research. In Eqs. (59)–(61), f_c, f_t, f_{tm} and E_c are in MPa and G_F in N/m. The parameters related to the yield surface in the concrete constitutive model were adopted as $\gamma = 0.01$, $\theta = 1.5$ and

$$\alpha = 1.2 + 0.1\sqrt{\frac{|f_c|}{f_t}}. \tag{62}$$

As for the elastic parameter to limit the elastic displacement in cohesive elements, it was assumed as $\zeta = 10$ (see also Ref. [46]).

In the rebar–concrete contact model employed in this research (Fig. 12), the maximum and residual strengths, as well as the exponent

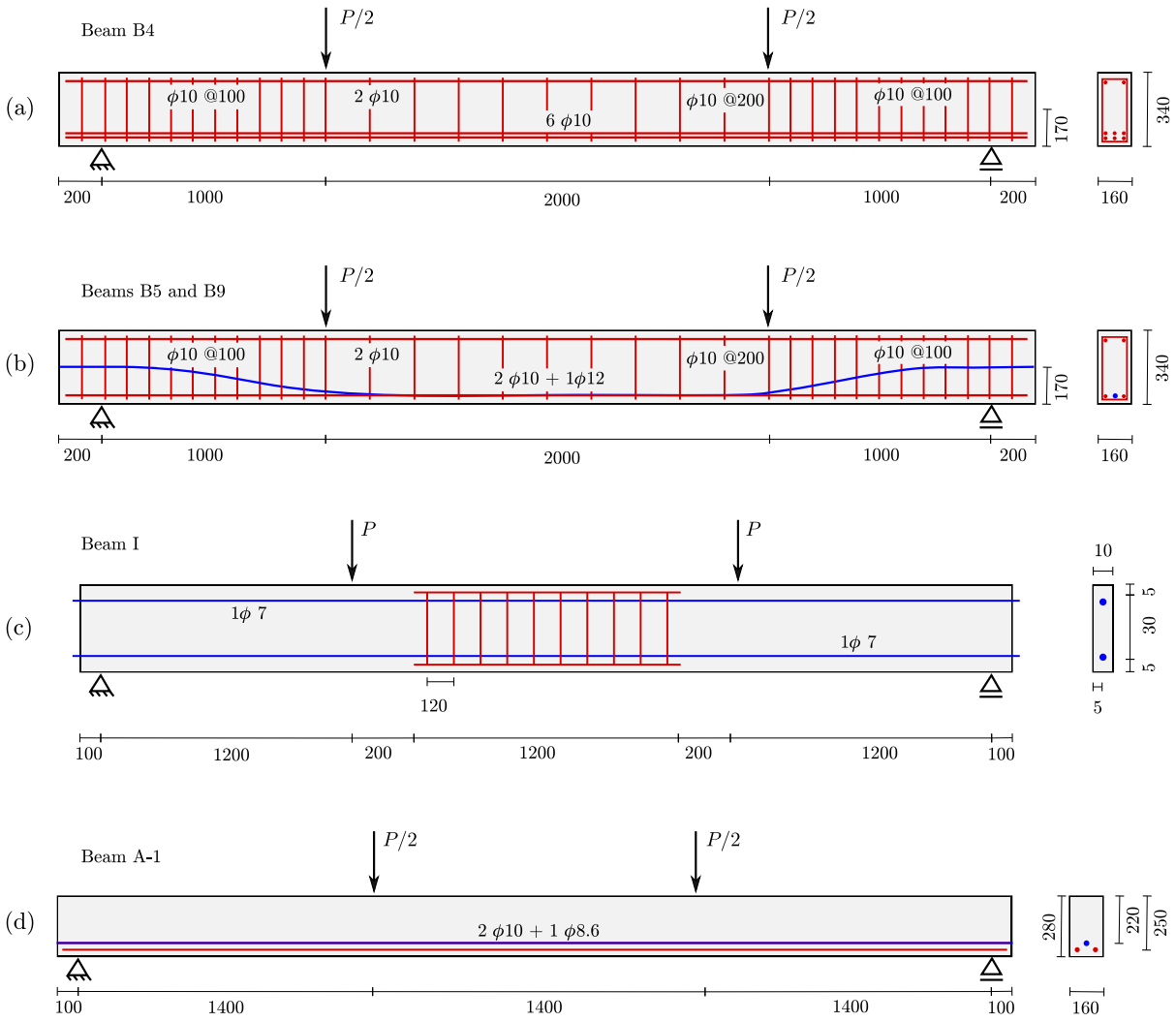


Fig. 15. Geometry, reinforcement and boundary conditions of simulated beams: (a) Non prestressed beam B4 by Hussien et al. [87]; (b) Prestressed beams B5 and B9 (bonded and unbonded, respectively) by Hussien et al. [87]; (c) Prestressed beam I by Eibl et al. [88]; (d) Prestressed beam A-1 by Tao and Du [89]. Blue reinforcements represent prestressing tendons.

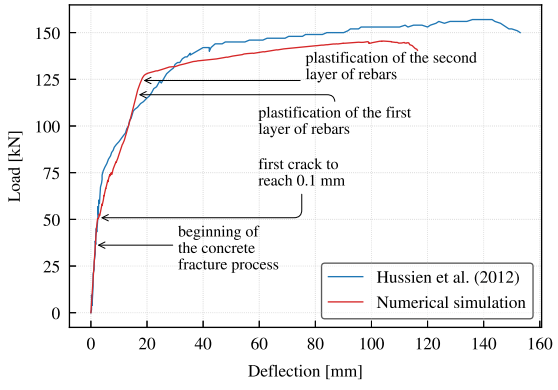


Fig. 16. Numerical and experimental results for beam B4.

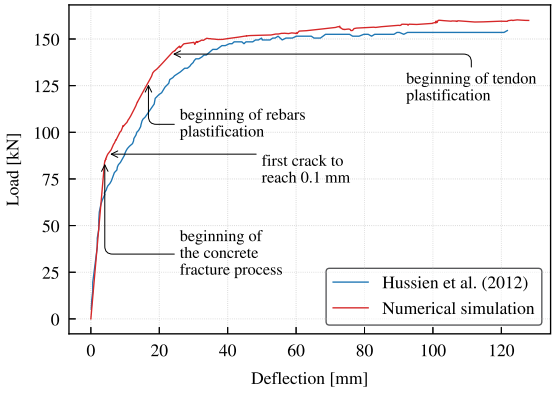


Fig. 17. Numerical and experimental results for beam B5.

from the ascending branch, were estimated according to CEB-FIP [90] code as

$$\tau_{max} = 2.5\sqrt{|f_c|} \tag{63}$$

$$\tau_{res} = 0.4\tau_{max} \tag{64}$$

$$\alpha = 0.4 \tag{65}$$

The peak slip was adopted as $s_{peak} = 0.07\phi_s$ (see Ref. [70]) where ϕ_s is the rebar diameter. In turn, s_{res} was estimated as the clear distance

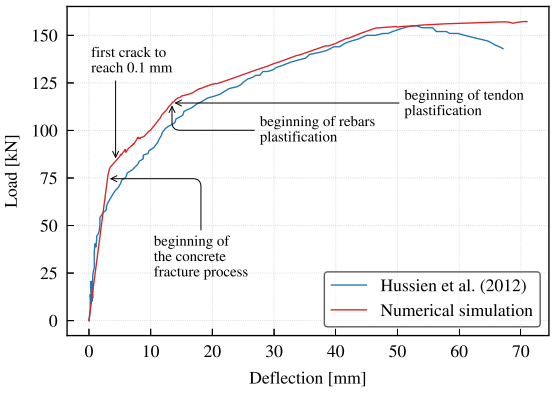


Fig. 18. Numerical and experimental results for beam B9.

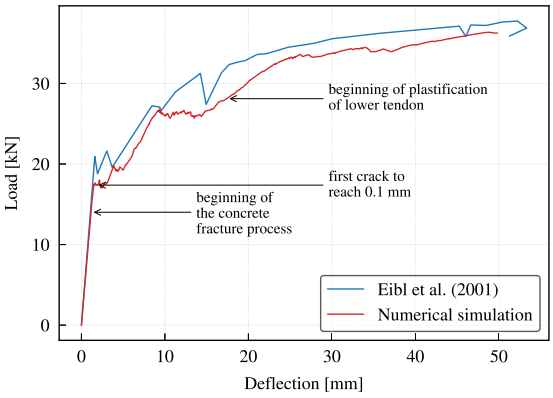


Fig. 19. Numerical and experimental results for beam I.

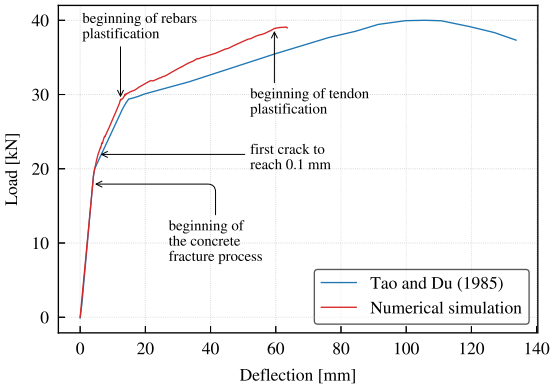


Fig. 20. Numerical and experimental results for beam A-1.

between rebar ribs, which is approximated by $0.4\phi_s$. The elastic parameters were adopted as $k_s = 1.5\frac{\tau_{max}}{s_{peak}}$ and $k_n = 10^3$ GPa/m. Tables 3–5 present a summary of the material parameters used in the simulations.

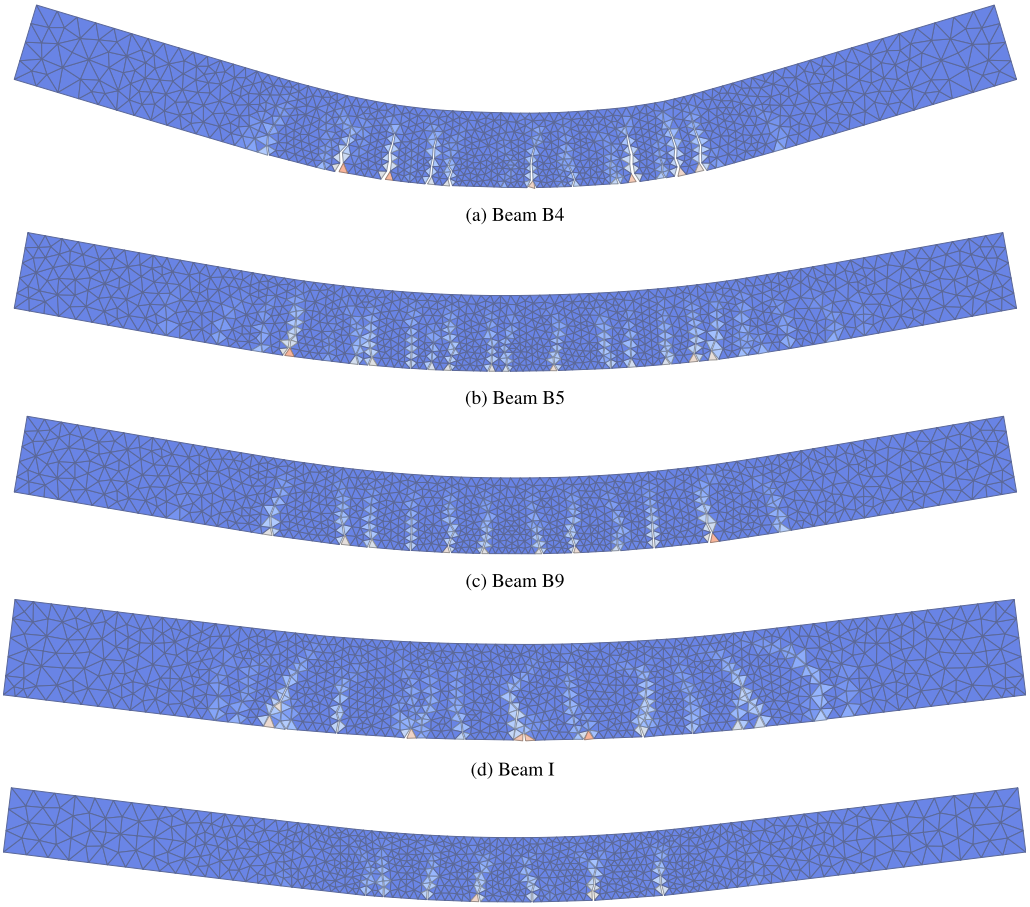
The numerical failure modes for all beams were similar. For instance, beam B4 failed due to the yielding of the steel reinforcement, followed by concrete crushing at the top of the beam. For the prestressed beams, failure occurred due to the yielding of the non-prestressed steel, followed by the yielding of the tendon. Figs. 16 through 19 compare the load–deflection curves obtained through numerical simulations and experimental tests. Overall, good agreement is observed in all cases. The numerical curves highlight some noteworthy points, such as the fracture process's onset and reinforcements' plastification. For the non-prestressed beam B4, a good agreement is observed regarding the overall behavior. An excellent agreement is also observed for prestressed beams B5 (bonded tendon) and B9 (unbonded tendon). The numerical and experimental results from beam I (pre-tensioned beam with bonded tendon) show many irregularities. However, this behavior is compatible with a shear failure mode which is corroborated by the crack pattern observed in Fig. 21(d). As for beam A-1 (post-tensioned beam with unbonded tendon), a good agreement can be observed in the curve shape (see Fig. 20). Although the numerical peak load is presented for a lower deflection, the peak values are close. In summary, the numerical curves exhibit similar shapes to the experimental curves, and the maximum loads are comparable in all simulations.

Fig. 21 presents the deformed beams at the end of each simulation using a scale factor of 3.5. To highlight the cracks, the figure also displays the plastic opening u_p in cohesive elements. Note that it is possible to see actual separations between bulk elements in some cases. Fig. 22 shows the crack patterns obtained at the end of the experiments. Very good agreement can also be observed in the crack patterns when comparing the numerical and experimental results. For example, for the non-prestressed beam B4, the numerical and experimental patterns show regularly distributed cracks along the beam. In turn, the cracks are more spaced for prestressed beams B5 and B9, especially in beam B9. For beam I, more cracks can be visualized in the numerical simulation, although the patterns are compatible. In this case, the experimental image only presented more visible cracks. Finally, for beam A-1, the numerical crack patterns are similar to the experiment since they are regularly distributed and concentrated near the center of the beam.

The proposed simulation approach can predict the discontinuity in the reinforcement contact forces due to cracks. For example, Fig. 23(a) shows the contact stress at reinforcements in beam B5. Strong discontinuities happen at crack locations for both the passive reinforcement and the bonded tendon. Conversely, this behavior is different in beam B9, as shown in Fig. 23(b). Since the tendon is unbonded, the discontinuities are only visible at the passive reinforcement. The proposed approach also allows for predicting axial stress variations in bonded and unbonded tendons. For example, the bonded tendon in beam B5 (see Fig. 24(a)) exhibits a non-uniform stress distribution, as the forces are transmitted to the concrete along the path of the tendon. On the other hand, the stress at the tendon in beam B9 (Fig. 24(b)) is practically constant since it is unbonded.

To further investigate the behavior of bonded and unbonded tendons in prestressed beams (B5 and B9, respectively), an analysis of tendon–concrete contact stresses and tendon axial stresses is conducted for different load levels. Fig. 25 displays the contact stress distributions, where sudden sign changes in the bonded tendon's contact stress correspond to the locations of cracks. As expected, the number of cracks increases with increasing load levels. On the other hand, the values of contact stress for the unbonded tendon are significantly lower than those for the bonded tendon. Fig. 26 displays the axial stress distributions along the tendons, where peaks in the bonded tendon's stress distribution coincide with the crack locations, whereas the stress distributions for the unbonded tendon tend to remain constant. Finally, to examine the crack distribution, Fig. 27 shows the horizontal displacements u_x at the base of the beams. These figures allow for easy estimation of the location and width of the cracks, as indicated by the sudden changes in the displacement distributions. For instance, note that the beam with the bonded tendon exhibits more concentrated distributions, consistent with the experiments.

In summary, the numerical results presented good agreement compared to the experiments. Some discrepancies in the results can be originated mainly from four different sources. First, some parameters required in the constitutive models are not available in the experimental works; thus, they were approximated using formulas proposed by design codes and previous research. Second, as it is known, concrete parameters obtained from samples may vary from the actual parameters in the structure due to, e.g., vibration issues, changes in aggregate



(a) Beam B4

(b) Beam B5

(c) Beam B9

(d) Beam I

(e) Beam A-1

Fig. 21. Deformed beams and crack patterns at the end of each simulation. The distribution of plastic displacements of cohesive elements is displayed to enhance the cracks' visualization.

Table 3
Material parameters for beams B4, B5 and B9.

Material/interface	Parameters			
Concrete	$E = 45$ GPa	$\nu = 0.2$	$f_c = 95$ MPa	$f_t = 6.5$ MPa
(B4)	$G_F = 166$ N/m	$\alpha = 1.58$	$\gamma = 0.01$	$\theta = 1.5$
Concrete	$E = 45$ GPa	$\nu = 0.2$	$f_c = 97$ MPa	$f_t = 4$ MPa
(B5 and B9)	$G_F = 166$ N/m	$\alpha = 1.69$	$\gamma = 0.01$	$\theta = 1.5$
Rebars	$E_s = 200$ GPa	$\sigma_y = 470$ MPa		
Tendon	$E_s = 200$ GPa	$\sigma_y = 1174$ MPa		
Rebars-concrete	$\tau_{max} = 24.4$ MPa	$\tau_{res} = 9.75$ MPa	$k_s = 52.2$ GPa/m	$k_n = 10^3$ GPa/m
contact	$s_{peak} = 0.7$ mm	$s_{res} = 4$ mm	$\alpha = 0.4$	$\beta = 0.4$
Tendon-concrete	$\tau_{max} = 24.6$ MPa	$\tau_{res} = 9.84$ MPa	$k_s = 43.7$ GPa/m	$k_n = 10^3$ GPa/m
contact (B5)	$s_{peak} = 0.84$ mm	$s_{res} = 4.8$ mm	$\alpha = 0.4$	$\beta = 0.4$
Tendon-concrete	$k_s = 10$ kPa/m	$k_n = 10^3$ GPa/m		
contact (B9)				
Rebars tips	$k_{tip} = 100$ GN/m			
Tendon tips	$k_{tip} = 100$ GN/m			

Table 4
Material parameters for beam I.

Material/contact	Parameters			
Concrete	$E = 26$ GPa	$\nu = 0.2$	$f_c = 44$ MPa	$f_t = 2.8$ MPa
	$G_F = 86$ N/m	$\alpha = 1.6$	$\gamma = 0.01$	$\theta = 1.5$
Rebars	$E_s = 180$ GPa	$\sigma_y = 470$ MPa		
Tendons	$E_s = 205$ GPa	$\sigma_y = 1420$ MPa		
Tendon-concrete	$\tau_{max} = 16.5$ MPa	$\tau_{res} = 6.6$ MPa	$k_s = 50.7$ GPa/m	$k_n = 10^3$ GPa/m
contact	$s_{peak} = 0.49$ mm	$s_{res} = 2.8$ mm	$\alpha = 0.4$	$\beta = 0.4$
Tendon tips	$k_{tip} = 100$ GN/m			

distribution, and discrepancies in curing time and method. Third, the experiments were performed only once for each beam setup; however, experimental results, such as RC loading tests, present noticeable variability among experiments even when tested under the same laboratory

Table 5
Material parameters for beam A-1.

Material/contact	Parameters			
Concrete	$E = 25$ GPa	$\nu = 0.2$	$f_c = 30.6$ MPa	$f_t = 2.0$ MPa
	$G_F = 46.6$ N/m	$\alpha = 1.6$	$\gamma = 0.01$	$\theta = 1.5$
Tendons	$E_s = 205$ GPa	$\sigma_y = 1465$ MPa		
Rebar–concrete contact	$\tau_{max} = 13.8$ MPa	$\tau_{res} = 5.5$ MPa	$k_s = 19.6$ GPa/m	$k_n = 10^3$ GPa/m
	$s_{peak} = 0.7$ mm	$s_{res} = 4$ mm	$\alpha = 0.4$	$\beta = 0.4$
Tendon-concrete contact	$k_s = 10$ kPa/m	$k_n = 10^3$ GPa/m		
Tendon tips	$k_{tip} = 100$ GN/m			

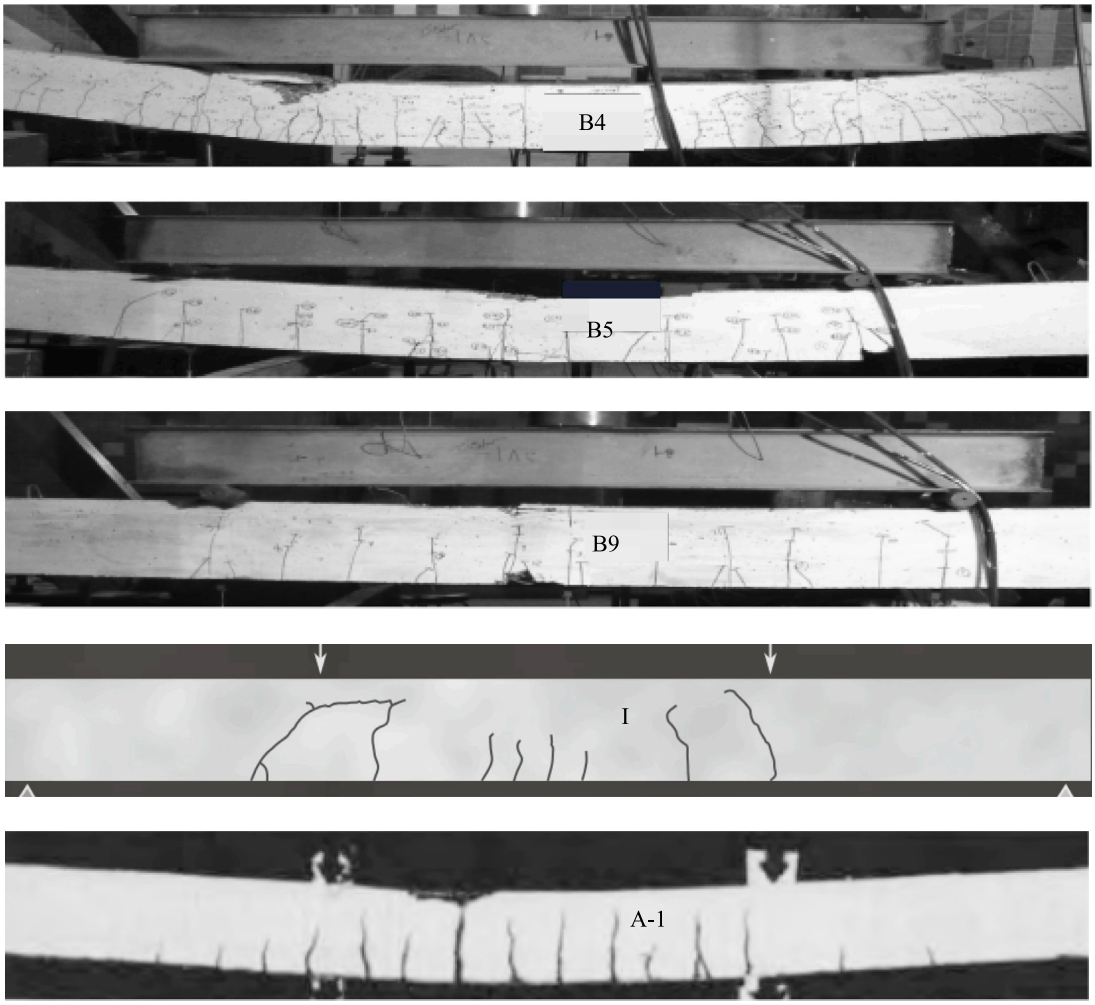


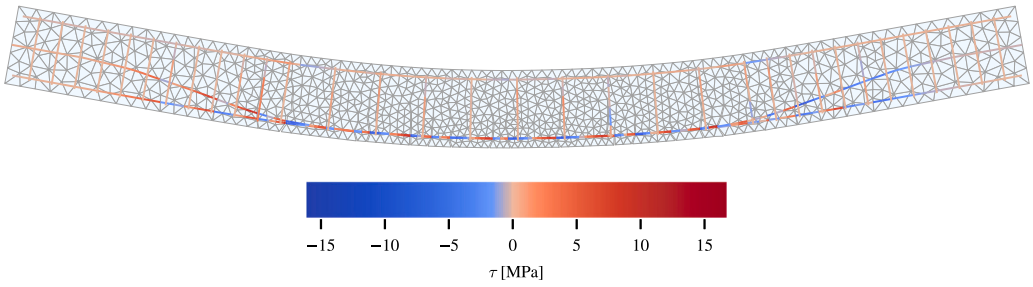
Fig. 22. Experimental crack patterns for beams B4, B5, B9 and A-1. Image for beam I was digitized from Ref. [91]. Remaining images were obtained from Refs. [87,89].

conditions. Finally, even the stress state in RC beams can be simplified to plane–stress conditions; there is some three-dimensional interaction between rebars (including stirrups) and concrete. Also, some cracks may be developed in the width direction. These features cannot be modeled using the two-dimensional concrete approach presented in this work.

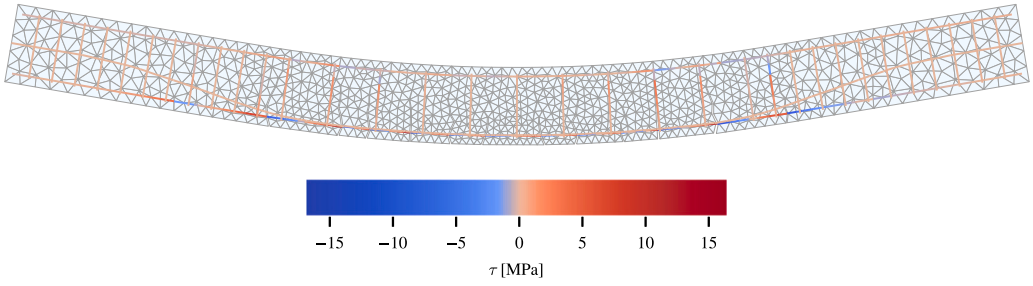
9. Conclusions

The paper presents a two-dimensional approach to simulate bonded and unbonded RC prestressed beams. This approach successfully combines several state-of-the-art finite element models to simulate the concrete (including fracture), tendons, passive reinforcements, and rebar–concrete contact. The study considers different phenomena, such as concrete cracking and anisotropy, rebar slip, rebar–concrete bond failure, and tip-bearing conditions. The approach also uses numerical techniques to improve the convergence of the loading process of the highly nonlinear problem. The proposed approach allows the effective simulation of RC prestressed beams, providing valuable information on deformability and cracking beyond rich details about the stress state in reinforcements.

Five experimental beams are simulated. Four are prestressed (pre- or post-tensioned) with bonded or unbonded tendons. The numerical results show very good agreement in the peak loads and the shapes of load–deflection curves. Similar to moment–curvature diagrams, these curves show well-defined loading phases as (1) linear-elastic, (2) onset of concrete cracking, (3) crack stabilization, and (4) steel-yielding. As for the numerical crack patterns, their distributions were consistent with the experimental results. In the numerical results, most cracks can easily be visualized by the detachment of bulk elements. Furthermore, the plastic opening in cohesive elements is displayed across the domain

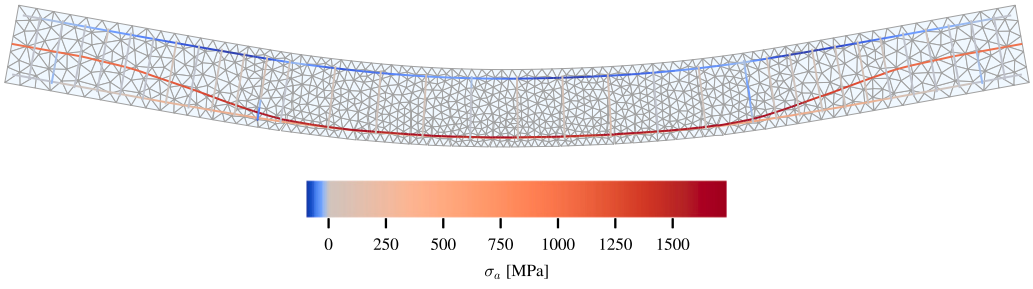


(a) Beam B5 (bonded).

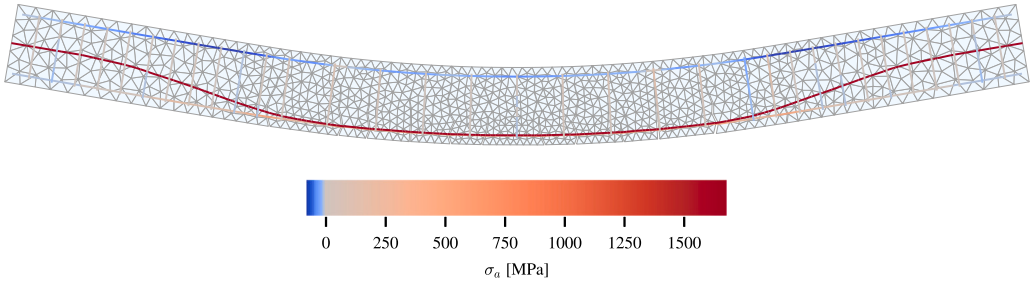


(b) Beam B9 (unbonded).

Fig. 23. Distribution of shear stress between rebars and concrete.



(a) Beam B5 (bonded).



(b) Beam B9 (unbonded).

Fig. 24. Distribution of reinforcements axial stress.

to improve the cracks' visualization. Additional results, such as axial and contact stress distributions at rebars and tendons, are consistent with bonded and unbonded prestressed beams. In particular, the proposed approach can predict contact stress discontinuities caused by cracks at passive rebars and bonded tendons. Finally, the authors envision a future extension to simulate RC structures in three-dimensional conditions.

CRediT authorship contribution statement

R. Durand: Conceptualization, Methodology, Resources, Software, Validation, Writing – original draft, Writing – review & editing. **J.F. Vieira:** Bibliographical review. **M.M. Farias:** Reviewing.

Declaration of competing interest

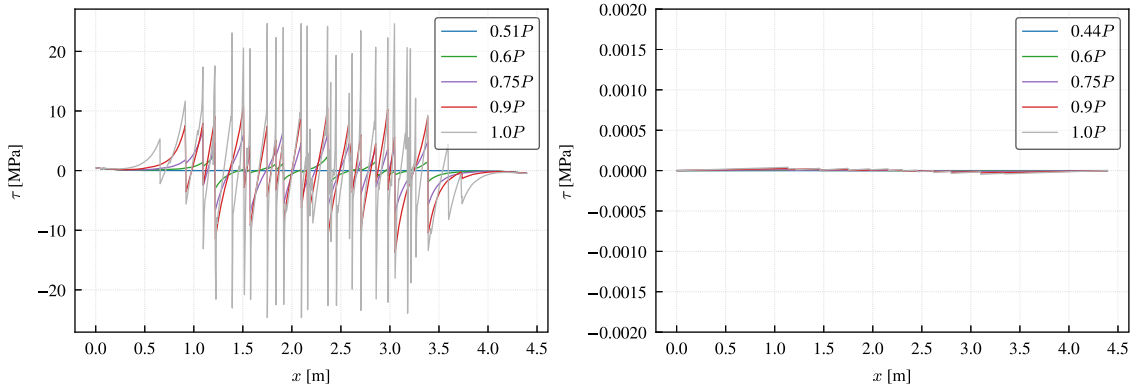
The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Raul Durand reports financial support was provided by Furnas Electricals.

Data availability

Data will be made available on request.

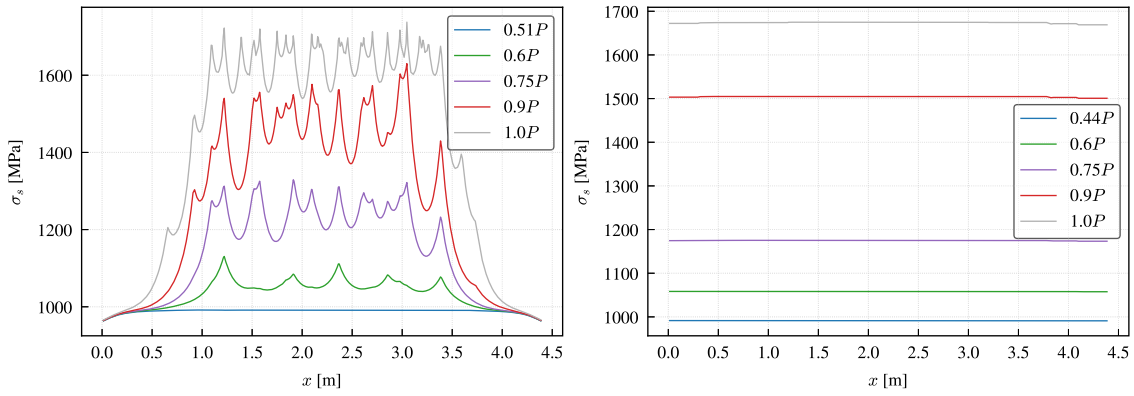
Acknowledgment

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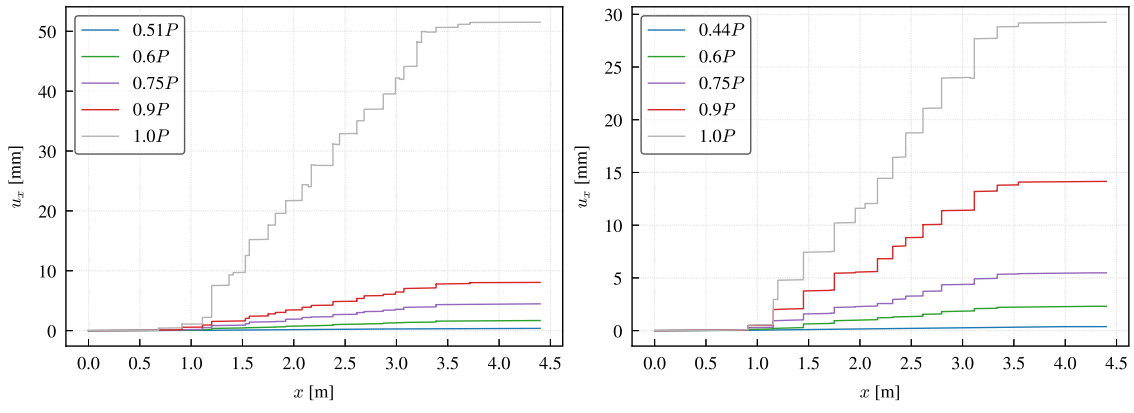
(a) Beam B5 (bonded). (b) Beam B9 (unbonded).

Fig. 25. Distributions of tendon–concrete contact stress for different load levels.



(a) Beam B5 (bonded). (b) Beam B9 (unbonded).

Fig. 26. Distributions of tendon axial stress for different load levels.



(a) Beam B5 (bonded). (b) Beam B9 (unbonded).

Fig. 27. Distribution of horizontal displacement u_x at the base of beams B5 and B9 for different load levels.

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